M5Stack Chain-PIR Protocol

All packets start with 0xAA 0x55 and end with 0x55 0xAA

V1 (Version) 2026/4/3

Command Set	Byte / Command Format	0	1	2	3	4	5	6	7	8	9	10
		Length_low	**Length_high**	**Index**	**Cmd**	**Data1**	**Data2**	**Data3**	**Data4**	**Data5**	**Data6**	**Data7**
Set RGB Values	Command Packet Format	0x05+Num*3	0x00	Index_id	0x20	Index	Num	R	G	B	CRC	
	Response Packet Format	0x04	0x00	Index_id	0x20	Operation_Status	CRC					
	Command Details	(1) Function: Set the RGB values (2) Input parameters: Index_id (device index ID), Index (RGB LED start index), Num (number of RGB LEDs to set), RGB value (3) Return parameter: Operation_Status () Command code: 0x20 Note 1: Operation_Status operation status 0: Operation failed 1: Operation successful Note 2: Index is the RGB LED start index and starts from 0. The Switch device has only one LED, so Index = 0 and Num = 1										
Get RGB Values	Command Packet Format	0x05	0x00	Index_id	0x21	Index	Num	CRC				
	Response Packet Format	0x04+Num*3	0x00	Index_id	0x21	Operation_Status	R	G	B	...	CRC	
	Command Details	(1) Function: Get the RGB values (2) Input parameters: Index_id (device index ID), Index (RGB LED start index), Num (number of RGB LEDs to read) (3) Return parameters: Operation_Status, RGB value (4) Command code: 0x21 Note 1: Operation_Status operation status 0: Operation failed 1: Operation successful Note 2: Index: RGB LED start index, starting from 0. The Switch device has only one LED, so Index = 0 and Num = 1										
Set RGB LED Brightness	Command Packet Format	0x05	0x00	Index_id	0x22	Light	Save_to_flash	CRC				
	Response Packet Format	0x04	0x00	Index_id	0x22	Operation_Status	CRC					
	Command Details	(1) Function: Set the RGB LED brightness (2) Input parameters: Index_id (device index ID), Light (RGB brightness), Save_to_flash (whether to save to internal Flash) (3) Return parameter: Operation_Status (4) Command code: 0x22 Note 1: Operation_Status operation status 0: Operation failed 1: Operation successful Note 2: Save_to_flash indicates whether to save to internal Flash 0: Do not save 1: Save Note 3: The setting takes effect immediately after being set successfully Note 4: Saving to internal Flash requires erasing a page. This process is time-consuming, and frequent operations will affect the device lifespan. If the brightness needs to be adjusted frequently, avoid saving to internal Flash every time.										
Get RGB LED Brightness	Command Packet Format	0x03	0x00	Index_id	0x23	CRC						
	Response Packet Format	0x04	0x00	Index_id	0x23	Light	CRC					
	Command Details	(1) Function: Get the RGB LED brightness. (2) Input parameter: Index_id (device index ID) (3) Return parameter: Light (RGB brightness) (4) Command code: 0x23										
Get PIR Detection Result	Command Packet Format	0x03	0x00	Index_id	0x37	CRC						
	Response Packet Format	0x04	0x00	Index_id	0x37	PIR_Value	CRC					
	Command Details	(1) Function: Get the PIR infrared detection result. (2) Input parameter: Index_id (device index ID) (3) Return parameter: PIR_Value value (4) Command code: 0x37 Note 1: PIR_Value (infrared detection result) 0: Person left 1: Person arrived											

PIR Detection Event Trigger	Command Packet Format	None										
	Response Packet Format	0x05	0x00	Index_id	0xE0	Status	0x05	CRC				
	Command Details	(1) Function: When the infrared detection event is triggered, the Chain PIR device automatically reports the current infrared detection result. (2) Input parameter: Index_id (device index ID) (3) Return parameter: Status (infrared detection result) (4) Command code: 0xE0 Note 1: Status infrared detection result 0: Person left 1: Person arrived Note 2: Infrared detection trigger condition: the event is triggered once whenever the PIR infrared detection result changes. Note 3: This packet is reported only when the "PIR Detection Event" trigger switch is enabled. After the packet is sent, the RGB LED overlays an effect on top of the user-configured lighting effect by adding to the RGB G value. Example 1: The user sets the RGB LED to (R=128, G=0, B=128) and the brightness to 50%. After brightness compression, the actual display is: R=64, G=0, B=64. When a PIR detection result change is triggered, 255 is added to the G channel, and the display becomes: R=64, G=255, B=64. Then the G value gradually decays to 0 within 1 second and finally returns to: R=64, G=0, B=64. Example 2: The user sets the RGB LED to (R=128, G=128, B=0) and the brightness to 50%. After brightness compression, the actual display is: R=64, G=64, B=0. When a PIR detection result change is triggered, 255 is added to the G channel. Because the original G value is 64, the overlaid value is 319. Values greater than 255 are capped at 255, so the display becomes: R=64, G=255, B=0. Then the G value gradually decays to 0 within 1 second and finally returns to: R=64, G=64, B=0. Note 4: 0x05 is the automatic report type code of the Chain PIR device.										
Set "PIR Detection Event Trigger" Switch	Command Packet Format	0x04	0x00	Index_id	0xE1	Mode	CRC					
	Response Packet Format	0x04	0x00	Index_id	0xE1	Operation_Status	CRC					
	Command Details	(1) Function: Set the switch for infrared detection event triggering. (2) Input parameters: Index_id (device index ID), Mode (PIR mode) (3) Return parameter: Operation_Status (4) Command code: 0xE1 Note 1: Mode (enable/disable event-triggered reporting, default: 1) 0: Trigger disabled 1: Trigger enabled Note 2: Operation_status operation status 0: Operation failed 1: Operation successful										
Get "PIR Detection Event Trigger" Switch	Command Packet Format	0x03	0x00	Index_id	0xE2	CRC						
	Response Packet Format	0x04	0x00	Index_id	0xE2	Mode	CRC					
	Command Details	(1) Function: Query the switch for infrared detection event triggering. (2) Input parameter: Index_id (device index ID) (3) Return parameter: Mode (enable/disable event-triggered reporting) (4) Command code: 0xE2 Note 1: Mode (report an event when a person is detected arriving/leaving, default: 1) 0: Trigger disabled 1: Trigger enabled										
Set "Person Arrived" Event Trigger Hold Time	Command Packet Format	0x05	0x00	Index_id	0xE3	Keep_Seconds	Save_to_flash	CRC				
	Response Packet Format	0x04	0x00	Index_id	0xE3	Operation_Status	CRC					
	Command Details	(1) Function: Set the hold time for the PIR infrared detection "person arrived" event trigger. (2) Input parameters: Index_id (device index ID), Keep_Seconds (hold seconds for the "person arrived" event trigger), Save_to_flash (whether to save to internal Flash) (3) Return parameter: Operation_Status (4) Command code: 0xE3 Note 1: Keep_Seconds ranges from 0 to 255 seconds, default: 5 seconds Note 2: Save_to_flash indicates whether to save to internal Flash 0: Do not save 1: Save Note 3: Operation_status operation status 0: Operation failed 1: Operation successful Note 4: The setting takes effect immediately after being set successfully Note 5: Saving to internal Flash requires erasing a page. This process is time-consuming, and frequent operations will affect the device lifespan.										

Get "Person Arrived" Event Trigger Hold Time	Command Packet Format	0x03	0x00	Index_id	0xE4	CRC						
	Response Packet Format	0x04	0x00	Index_id	0xE4	Keep_Seconds	CRC					
	Command Details	(1) Function: Get the hold time for the PIR infrared detection "person arrived" event trigger. (2) Input parameter: Index_id (device index ID) (3) Return parameter: Keep_Seconds (hold seconds for the "person arrived" event trigger) (4) Command code: 0xE4 Note 1: Keep_Seconds ranges from 0 to 255 seconds, default: 5 seconds										
Query Device Unique UID	Command Packet Format	0x04	0x00	Index_id	0xF8	UID_Type	CRC					
	Response Packet Format	0x04+4/12	0x00	Index_id	0xF8	Operation_status	UID (multi-byte)	CRC				
	Command Details	(1) Function: Query the device unique UID. (2) Input parameters: Index_id (device index ID), UID_Type (3) Return parameters: Operation_status, UID (4) Command code: 0xF8 Note 1: Operation_status operation status 0: Operation failed 1: Operation successful Note 2: UID_Type UID type 0: 4-byte UID 1: 12-byte UID										
Query Bootloader Version	Command Packet Format	0x03	0x00	Index_id	0xF9	CRC						
	Response Packet Format	0x04	0x00	Index_id	0xF9	Bootloader_version	CRC					
	Command Details	(1) Function: Query the bootloader version. (2) Input parameter: Index_id (device index ID) (3) Return parameter: Bootloader_version (4) Command code: 0xF9										
Query Device Firmware Version	Command Packet Format	0x03	0x00	Index_id	0xFA	CRC						
	Response Packet Format	0x04	0x00	Index_id	0xFA	Firmware_version	CRC					
	Command Details	(1) Function: Query the device firmware version. (2) Input parameter: Index_id (device index ID) (3) Return parameter: Firmware_version (4) Command code: 0xFA										
Query Device Type	Command Packet Format	0x03	0x00	Index_id	0xFB	CRC						
	Response Packet Format	0x05	0x00	Index_id	0xFB	Device_type_low	Device_type_high	CRC				
	Command Details	(1) Function: Query the device type. (2) Input parameter: Index_id (device index ID) (3) Return parameter: Device_type (4) Command code: 0xFB Note 1: Device_type = (uint16_t)((Device_type_high << 8) | Device_type_low)										
Enumeration Request	Command Packet Format	None										
	Response Packet Format	0x03	0x00	Index_id	0xFC	CRC						
	Command Details	(1) Function: Enumeration request. Sent by the terminal device when the chain link changes and sent when the device powers on, notifying the host to update the chain device status. (2) Input parameter: none (3) Return parameter: none (4) Command code: 0xFC										
Heartbeat Packet	Command Packet Format	0x03	0x00	0xFF	0xFD	CRC						
	Response Packet Format	0x03	0x00	0xFF	0xFD	CRC						
	Command Details	(1) Function: Heartbeat packet. Chain devices communicate periodically to determine whether they are the terminal device. The host can also use heartbeat packets to determine whether chain devices are connected. (2) Input parameter: none (3) Return parameter: none (4) Command code: 0xFD										

Enumerate	Command Packet Format	0x04	0x00	0xFF	0xFE	Send_num	CRC					
	Response Packet Format	0x04	0x00	0xFF	0xFE	Receive_num	CRC					
	Command Details	(1) Function: The user can obtain the number of devices in the chain through enumeration. (2) Input parameter: Send_num (default: 0, used to record the number of devices) (3) Return parameter: Receive_num (the value represents the number of devices) (4) Command code: 0xFE										

Note 1: The maximum packet length is 256 bytes for serial communication
Note 2: The data length is from Index_id to CRC, including Index_id and CRC, but excluding the data length field itself
Note 3: During CRC calculation, exclude the packet header, packet tail, length, and CRC fields themselves, and sum only the remaining data
Note 4: Serial communication baud rate: 115200; 8 data bits; 1 stop bit; no parity bit

```
uint8_t calculateCRC(const uint8_t *buffer, uint16_t size)
{
    uint8_t crc8 = 0;
    for (uint8_t i = 4; i < (size - 3); i++) {
        crc8 += buffer[i];
    }
    return crc8;
}
```